Security

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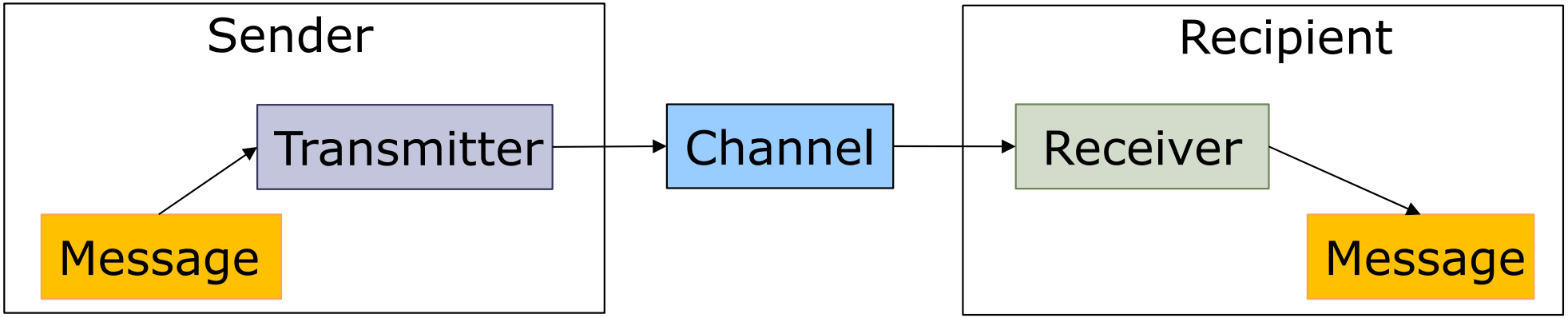
Security and Information Leakage

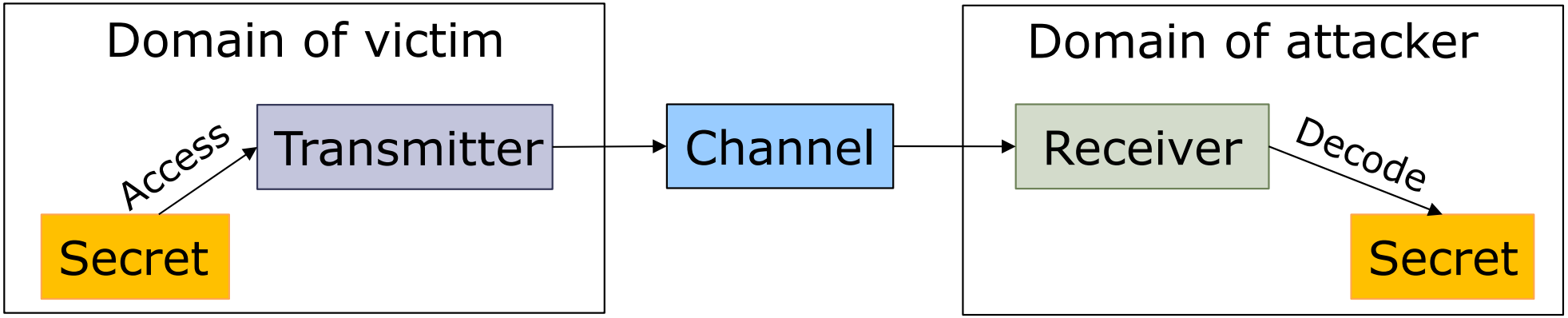
- Hardware isolation mechanisms like virtual memory guarantee that architectural state will not be directly exposed to other processes…but
- ISA is a timing-independent interface, and
  - Specify *what* should happen, not *when*
- ISA only specifies architectural updates (reg, mem, PC…)
  - *Micro-architectural changes are left unspecified*
- So implementation details and timing behaviors (e.g., microarchitectural state, power, etc.) have been exploited to breach security mechanisms.
- In specific, they have been used as channels to leak information!

Standard Communication Model



1. Transmitter gets a message
2. Transmitter modulates channel
3. Receiver detects modulation on channel
4. Receiver decodes modulation as message

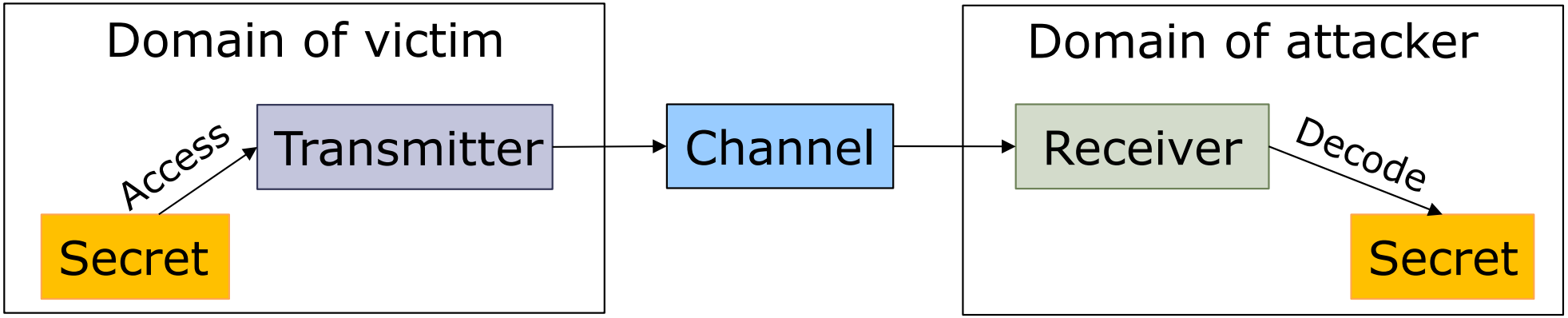
Communication Model of Attacks
[Belay, Devadas, Emer]



- Domains – Distinct architectural domains in which architectural state is not shared.
- Secret – the "message" that is transmitted on the channel and detected by the receiver
- Channel – some "state" that can be changed, i.e., modulated, by the "transmitter" and whose modulation can be detected by the "receiver".

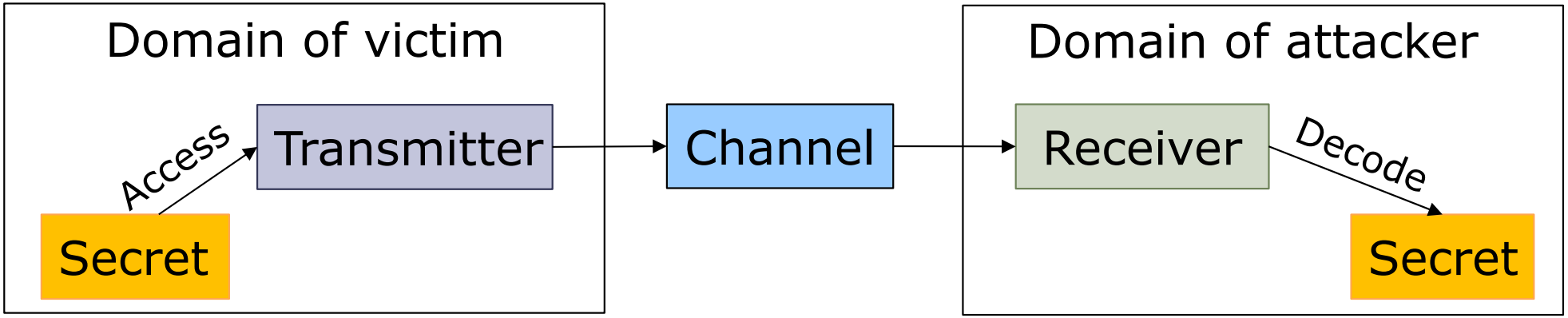
Because channel is not a "direct" communication channel, it is often referred to as a "side channel"

Communication Model of Attacks
[Belay, Devadas, Emer]



1. Transmitter "accesses" secret
2. Transmitter modulates channel *(microarchitectural state)* with a message based on secret
3. Receiver detects modulation on channel
4. Receiver decodes modulation as a message containing the secret

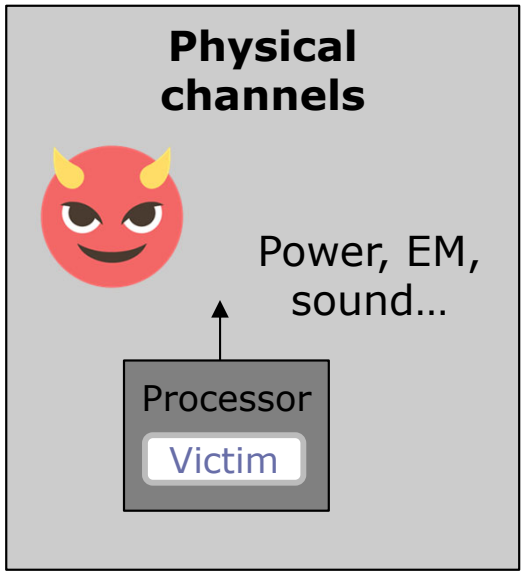
ATM Acoustic Channels



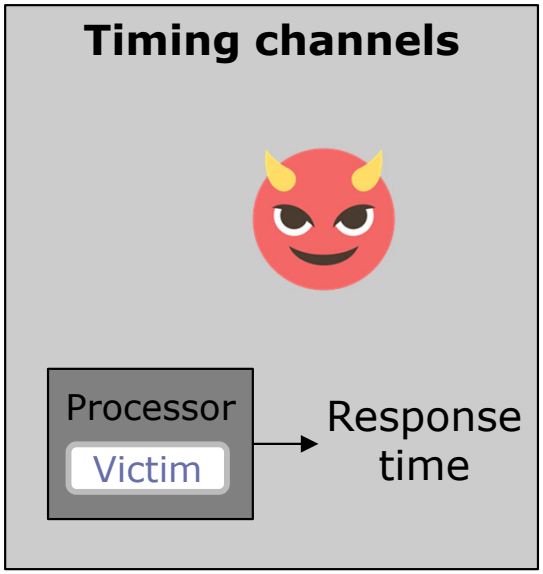
- Secret: Pin
- Transmitter: Keypad
- Channel: Air
- Modulation: Acoustic waves
- Receiver: Cheap Microphone
- Decoders: ML Model

Physical vs Timing vs uArch Channel

- Types of channels



Attacker requires measurement equipment → physical access



Attacker may be remote (e.g., over an internet connection)

What can you do with these channels?

- Violate privilege boundaries
  - Inter-process communication
  - Infer an application's secret
- (Semi-Invasive) application profiling

Different from traditional software or physical attacks:

- Stealthy. Sophisticated mechanisms needed to detect channel
- Usually, no permanent indication one has been exploited

Timing Channel Example

```
def check(input):

  size = len(passwd); //passwd contains 8 digits
  for i in range(0,size):
    if (input [i] != password[i]):
      return ("error");

  return (“success”)
```

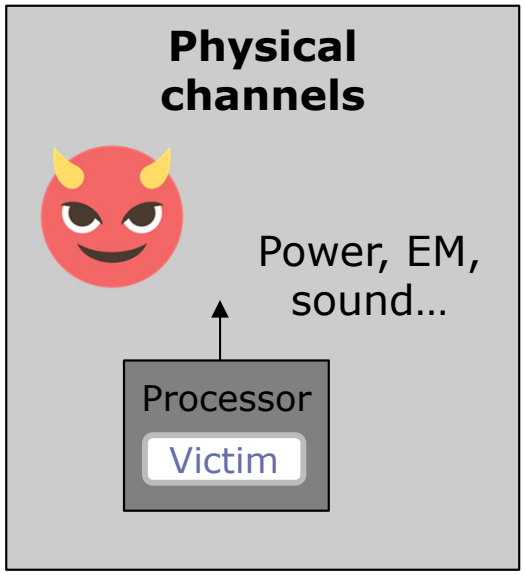
Blind guess needs to maximally try: 10^8

Can we do better to reduce the number of trials?

The execution time is dependent on how many characters match between the input and the correct password. Attacker can brute-force each character. Maximally try 10*8 times.

Physical vs Timing vs uArch Channel

- Types of channels



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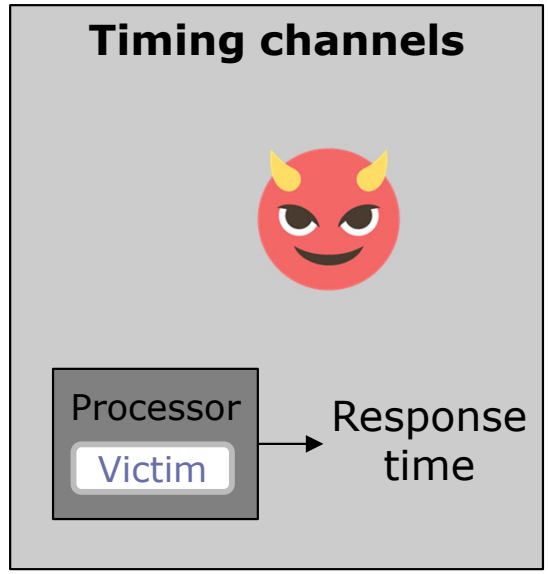

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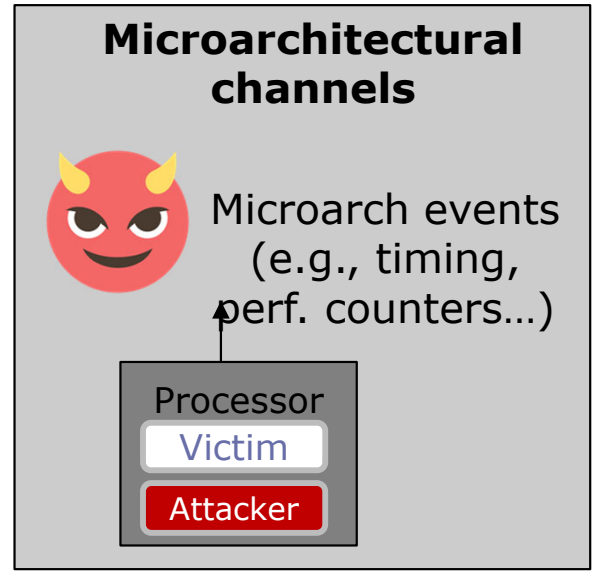
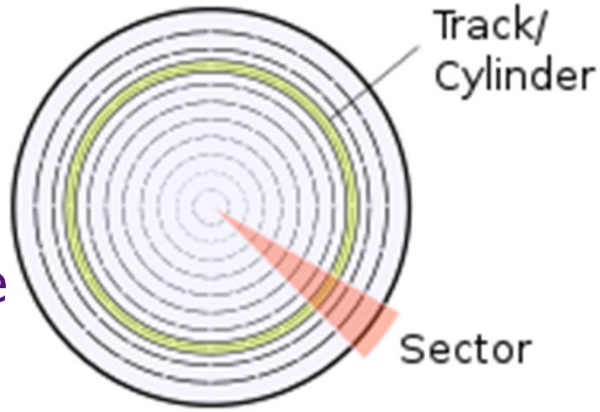


Attacker may be remote, or be co-located

Side Channel Attacks in 1977

- A side channel due to disk arm optimization
  - Enqueues requests by ascending cylinder number and dequeues (executes) them by the "elevator algorithm."



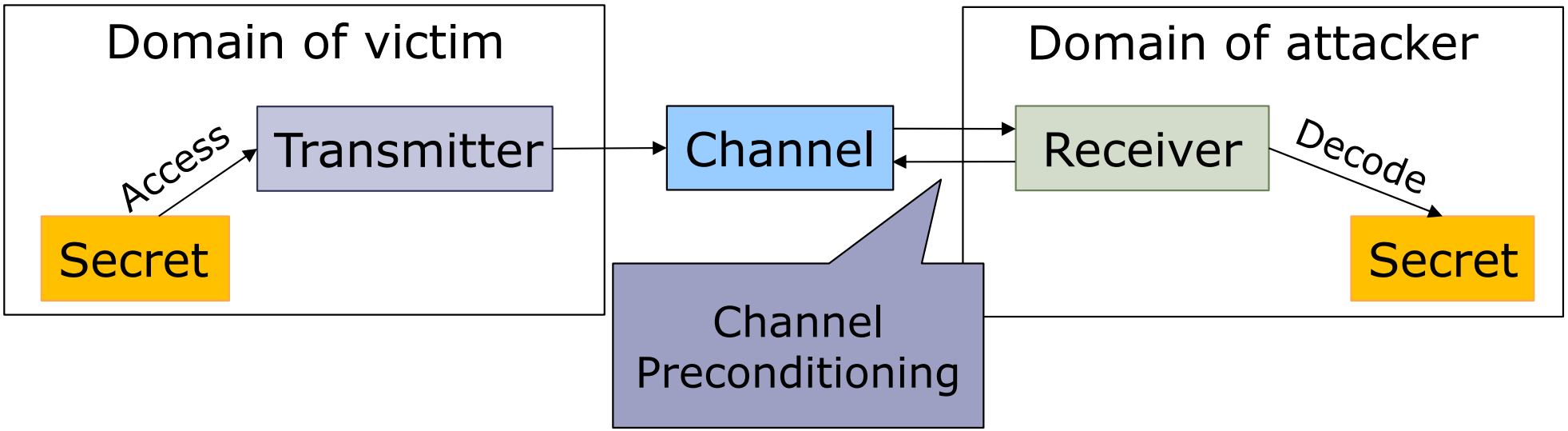
- Example:
  1. Receiver issues a request to 55 then pauses
  2. Transmitter issues a request to either 53 or 57
  3. Receiver pauses and issues requests to both 52 and 58

Q: If the Receiver receives data for 52 first, can we guess what did Transmitter issue before? 53

Q: If we remove step 1, can the attack still work? No

Note need for an "active" receiver that preconditions the channel

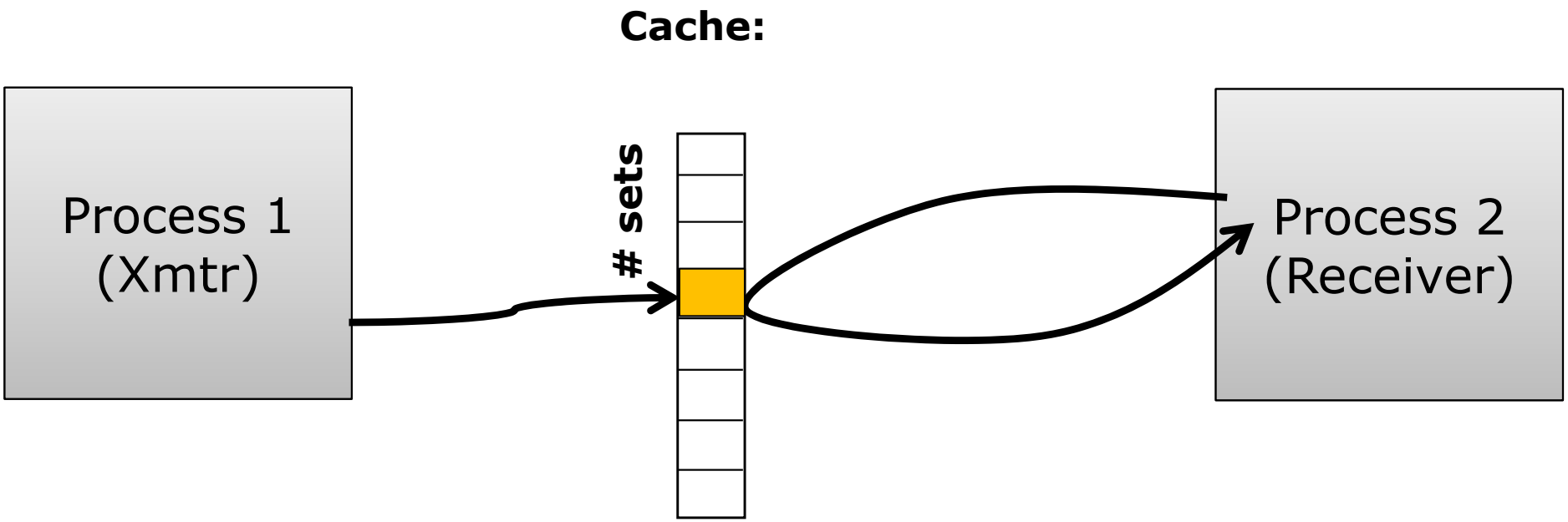
Communication w/ Active Receiver



- An active receiver may need to "precondition" the channel to prepare for detecting modulation

- An active receiver also needs to deal with synchronization of preconditioning, transmission (modulation) activity and reception (demodulation) activity.

A Cache-based Channel



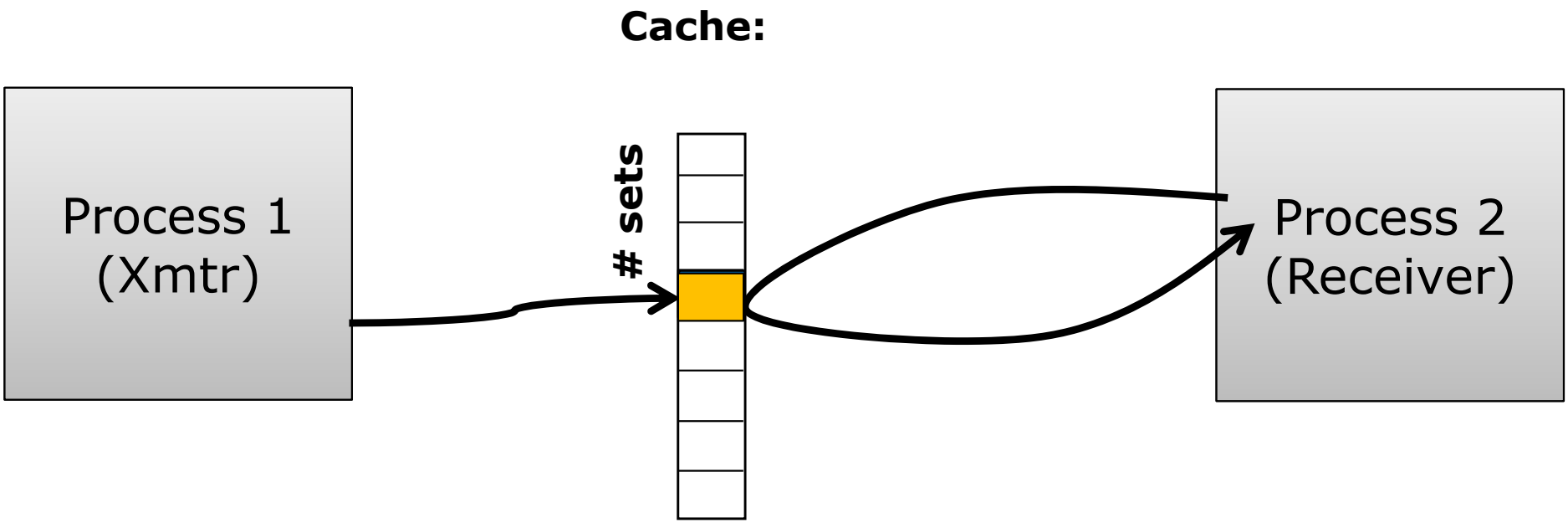
if (**send '0'**)
  idle
else
  write to a set

write to set

t1 = rdtsc()
read from the set
t2 = rdtsc()

if t2 – t1 > hit_time:
  decode '1'
else
  decode '0'

A Cache-based Channel



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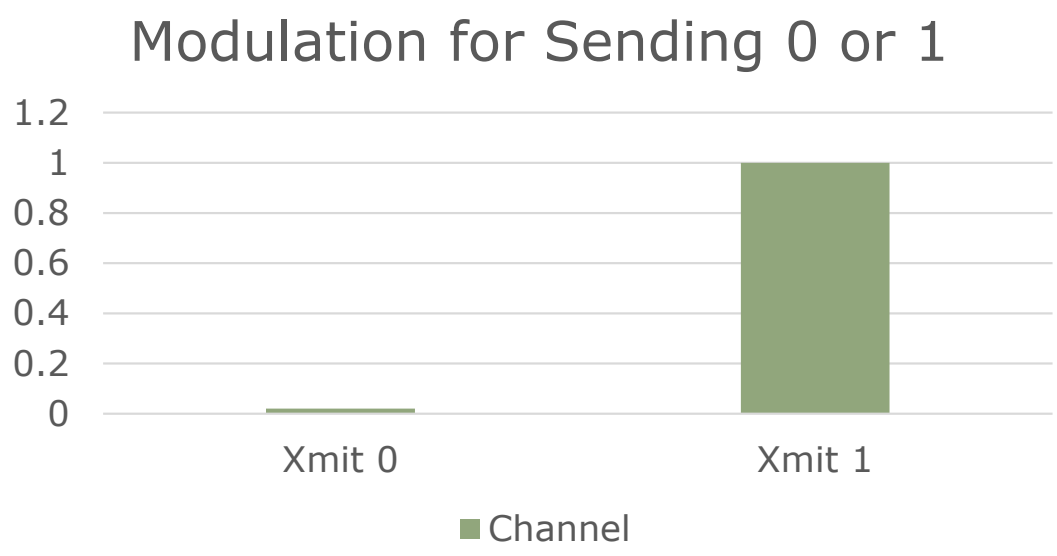
Simple Transmitter

```
secret = oneof(0..1)
if secret == 1:
    x = channel
```



Like an amplitude modulated (AM) radio transmission

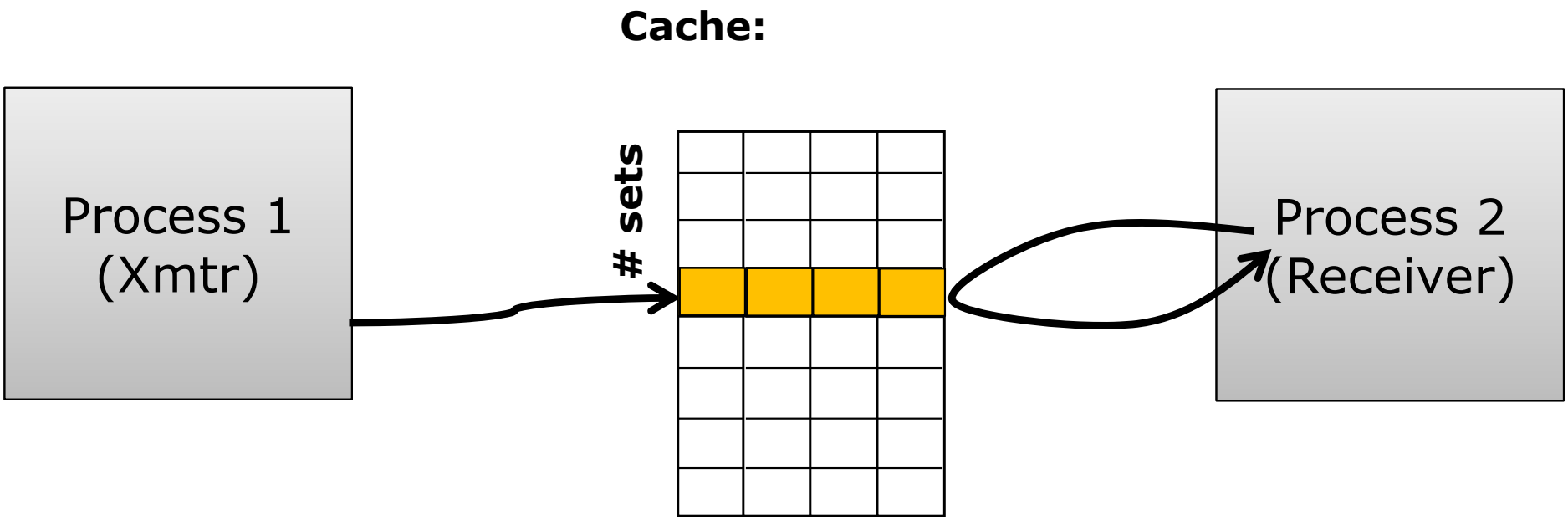
"AM" Transmitter in RSA
[Percival 2005]

- Square-and-multiply based exponentiation

Input : base b, modulo *m*,
exponent $e = (e_{n-1} \ldots e_0)_2$
Output: $b^e \bmod m$
$r = 1$
for $i = n-1$ *down to* 0 do
  $r = sqrt(r)$
  $r = mod(r,m)$
  if e_i **== 1 then**
    $r = mul(r,b)$
    $r = mod(r,m)$
  end
end
return r

Secret-dependent memory access → transmitter

A Multi-way Cache-based Channel



if (**send '0'**)
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fill a set

t1 = rdtsc()
read all of the set
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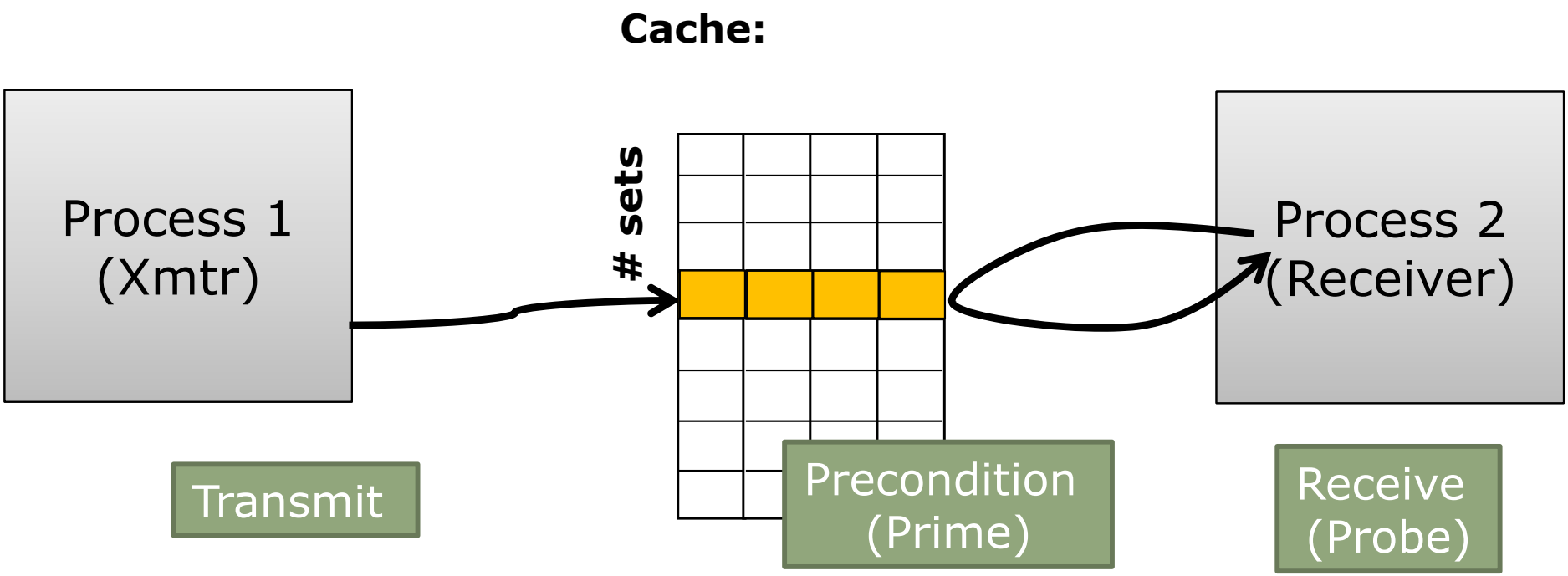
A Multi-way Cache-based Channel



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fill a set

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else
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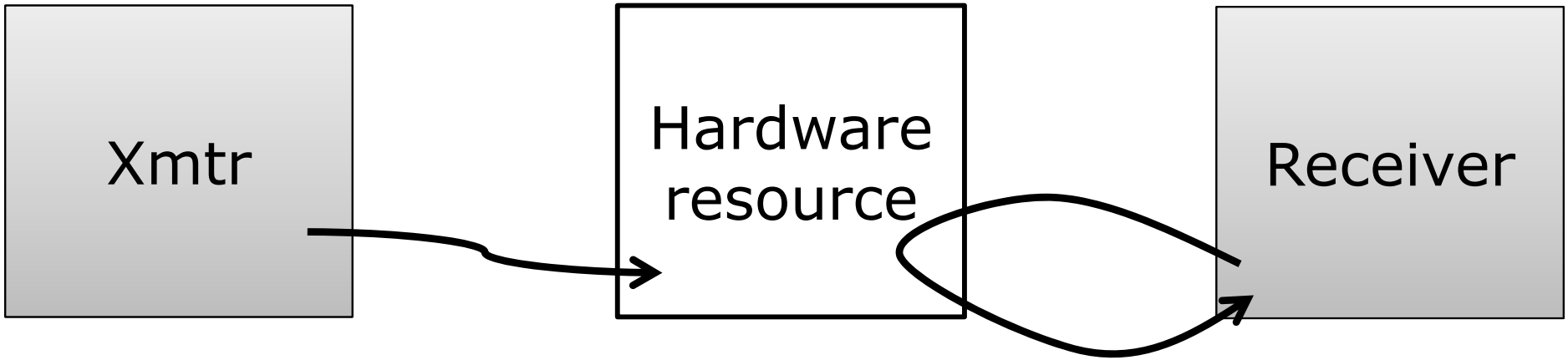
Decode

Generalizes to Other Resources




```
if (send '1')
  Use resource
else
  idle
```

```
t1 = rdtsc()
Use resource
t2 = rdtsc()
```

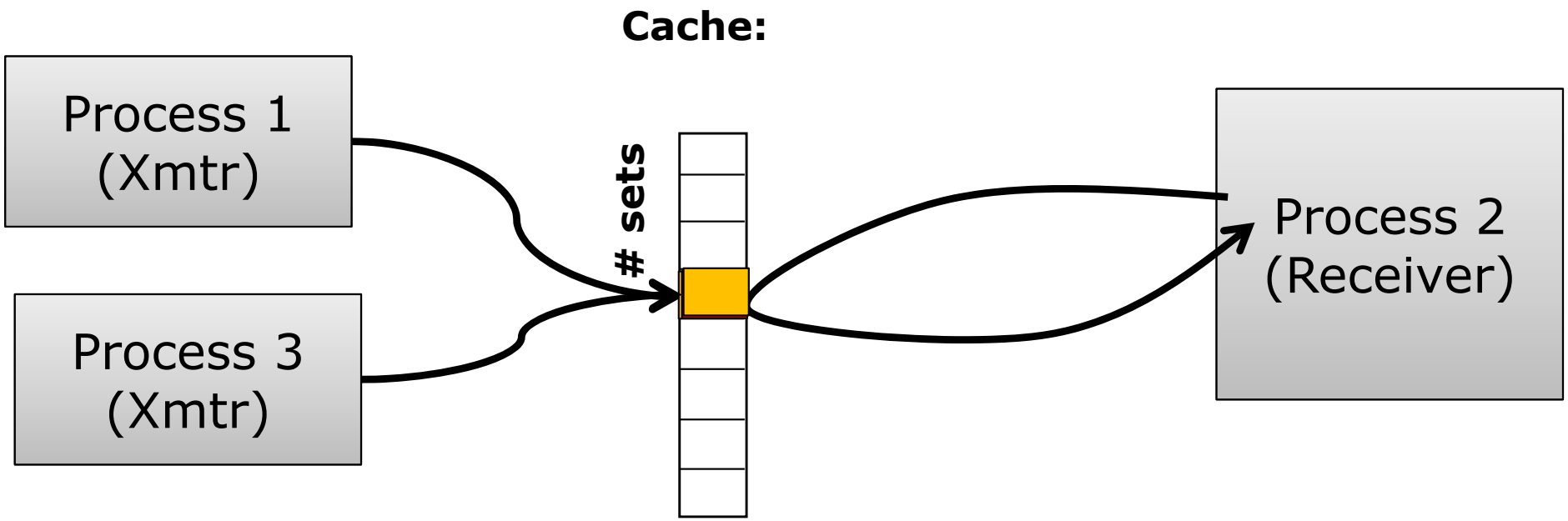
```
if (t2 – t1 > THRESH)
        read '1'
else
        read '0'
```

Any other exploitable structures?

Channel Examples

Resource	Shared by
Private cache (L1, L2)	Intra-core
Shared cache (LLC)	On-socket cross core
Cache directory	Cross socket
DRAM row buffer	Cross socket
TLB (private/shared)	Intra-core/Inter-core
Branch Predictor	Intra-core
Network-on-chip	On-socket cross core
…	…

Noise in the channel



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write to set

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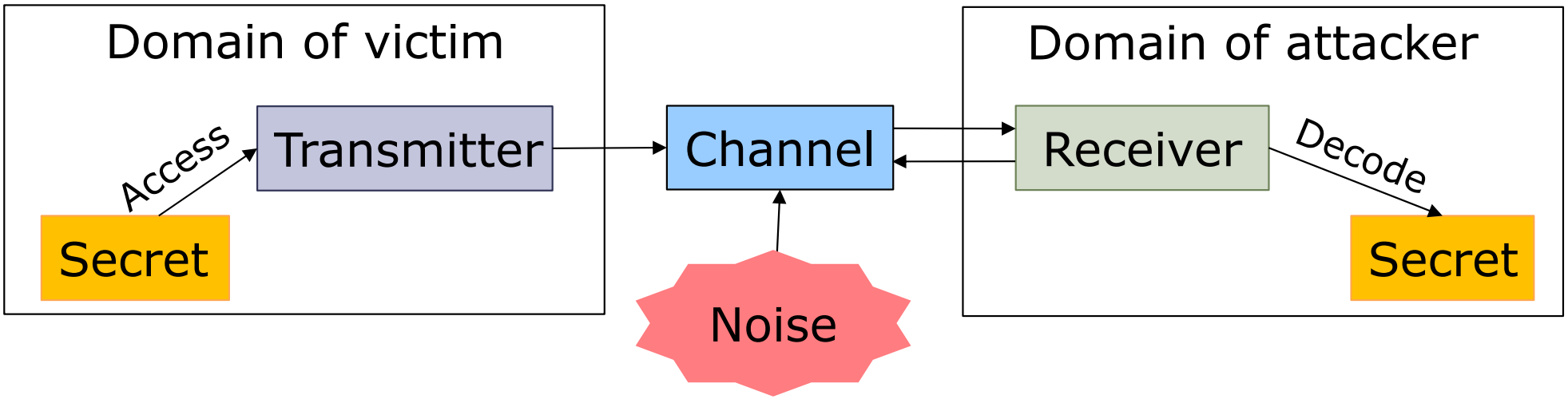
  decode '0'

Receiver interprets "noise" as a signal!

Channel Noise

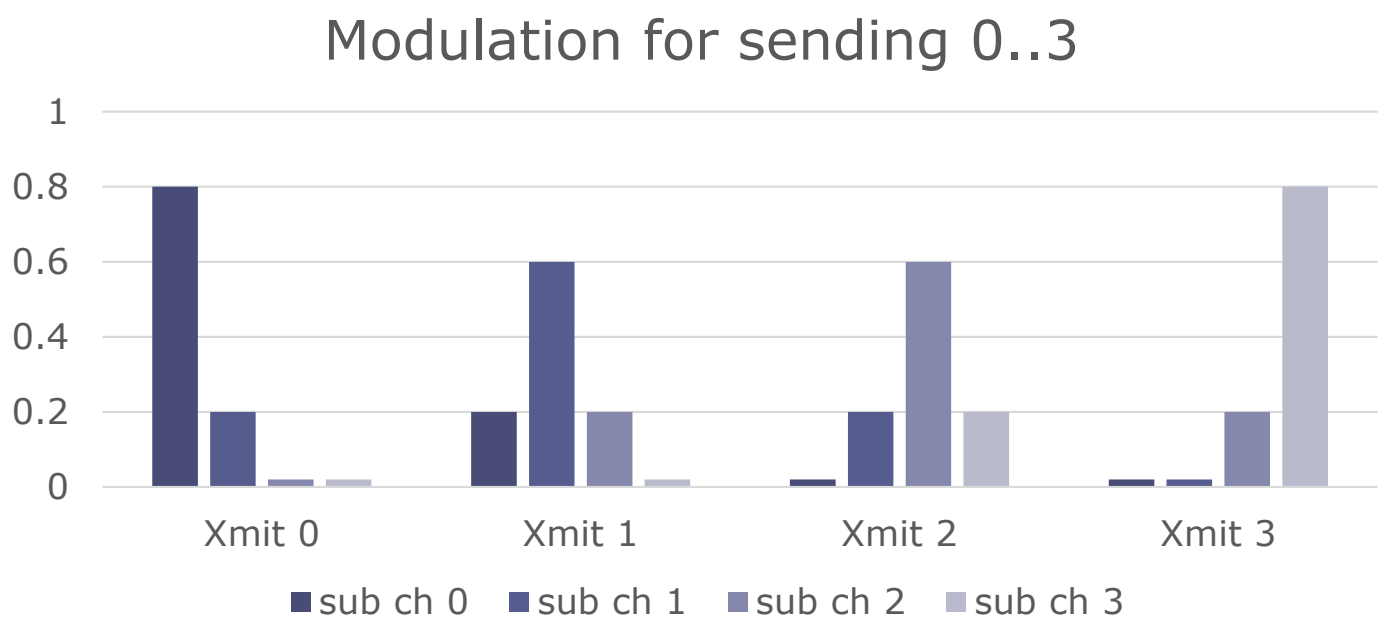


1. Another (or the same) transmitter may introduce changes of state (noise) into the channel which will confound the receiver
2. Reception now becomes probabilistic, and a stochastic analysis is needed for the receiver to decode the modulation it sees in the channel.
3. Increases in reliability of reception can be improved by improved message encoding, e.g., by repeating the message.

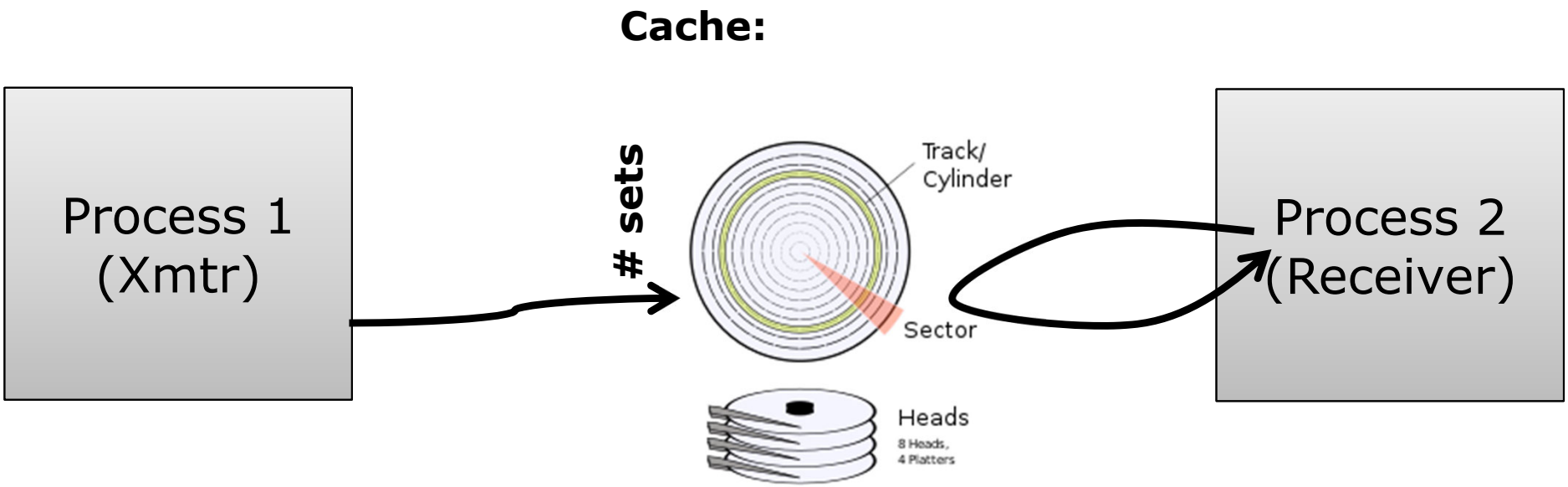
Noise makes signal probabilistic

secret* = *oneof(0..3)

subchannel[***secret***] = 1

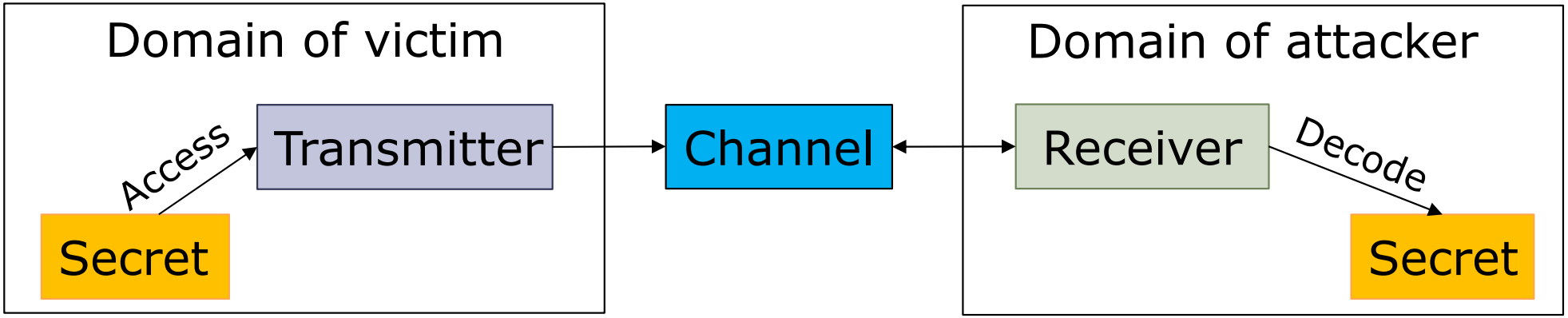


Disrupting Communication



"We found that identifying all of the sources of accurate clocks was much **easier** than finding all of the possible timing channels in the system.
… If we could make the clocks less accurate, then the effective bandwidth of all timing channels in the system would be **lowered**." *(1991)*

Secret-independent Channel Modulation



- Different from conventional communication, this is a side channel (*unintended* communication).

- One mitigation is to not use the channel.

-> ”data-oblivious execution” or “constant-time programming”.

Secret-independent Channel Modulation

Input : base **b**, modulo *m*,
exponent $\boldsymbol{e = (e_{n-1} \ldots e_0)_2}$
Output: $b^e \bmod m$

```
r = 1
for i = n−1 down to 0 do
  r = sqrt(r)
  r = mod(r,m)
    if e_i == 1 then
      r = mul(r,b)
      r = mod(r,m)
    end
end
return r
```

```
p = (e_i == 1)
r2 = mul(r,b)
r2 = mod(r,m)
cmov [p] r, r2
```

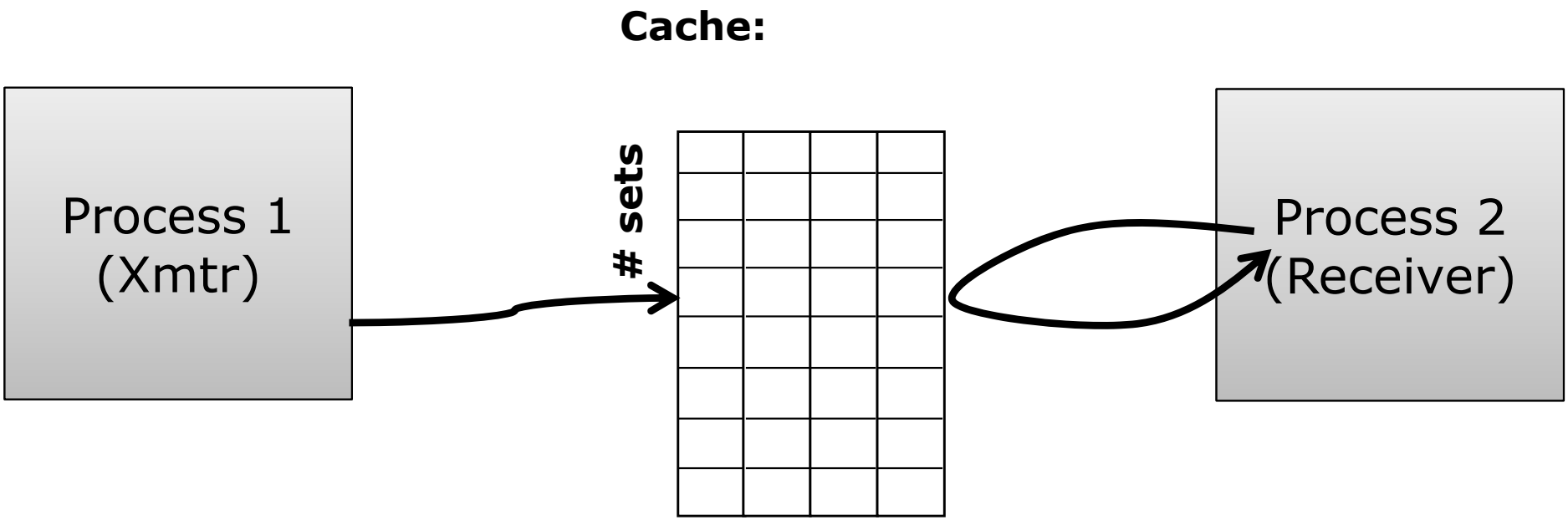
How to make the code execution independent of the secret?

No secret-dependent branches, memory accesses, floating point operations

After removing the secret-dependent branch, how about code inside these functions?

Constant-time programming is hard

Disrupting Communication



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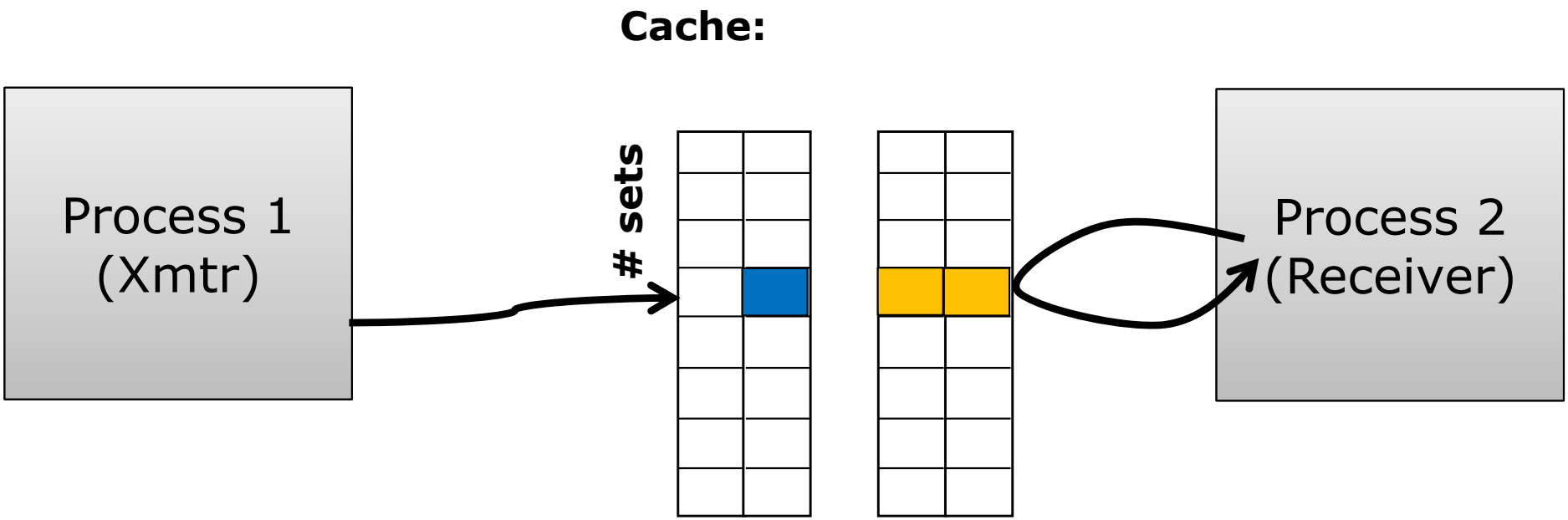

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fill a set

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read all of the set
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if t2 – t1 > hit_time:
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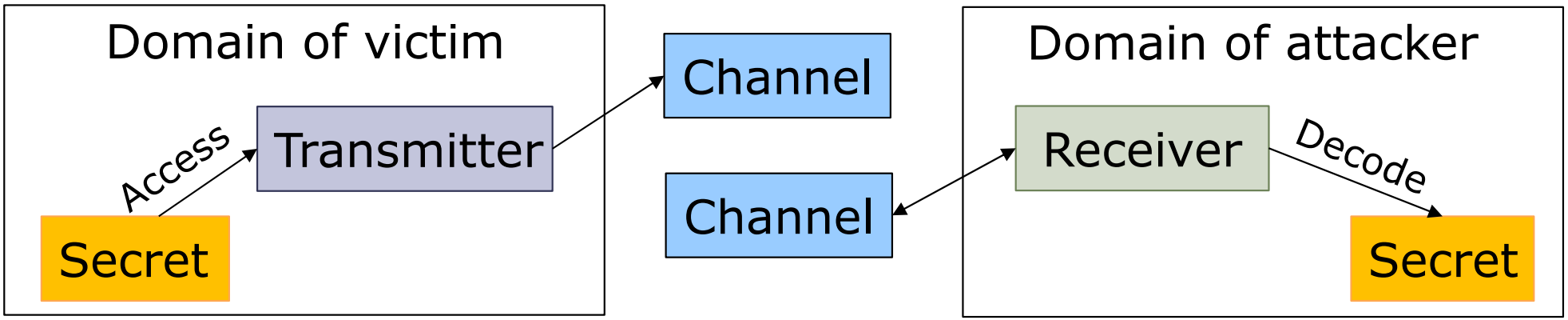
else
  decode '0'

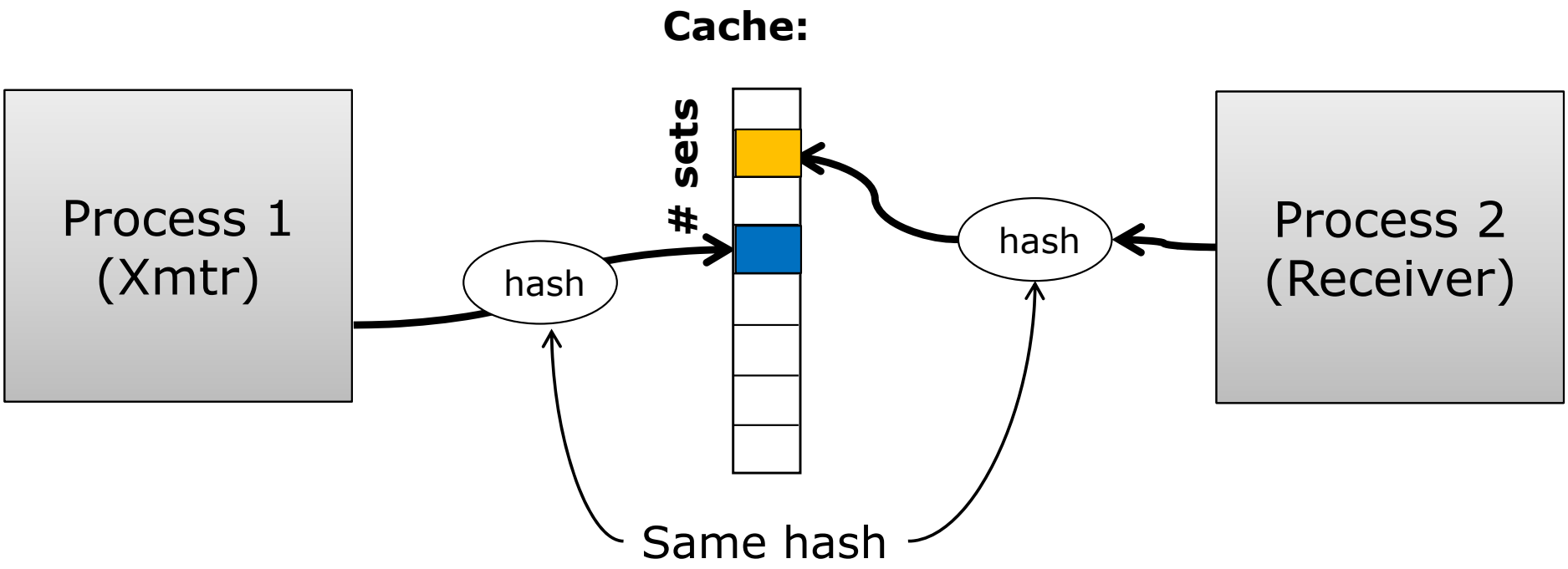
Kirianski et. al. Dawg, Micro'18

Disjoint Channels



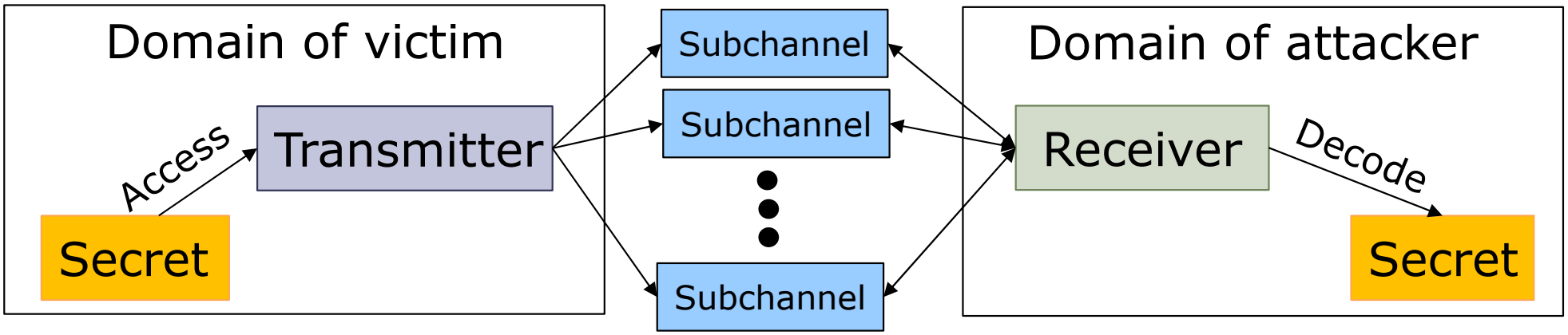
- Making disjoint channels makes communication impossible.
- Channel can be allocated by "domain" and will need to be "cleaned" as processes enter and leave running state, so next process cannot see any "modulation" on the channel.

Obfuscating the channel (1)



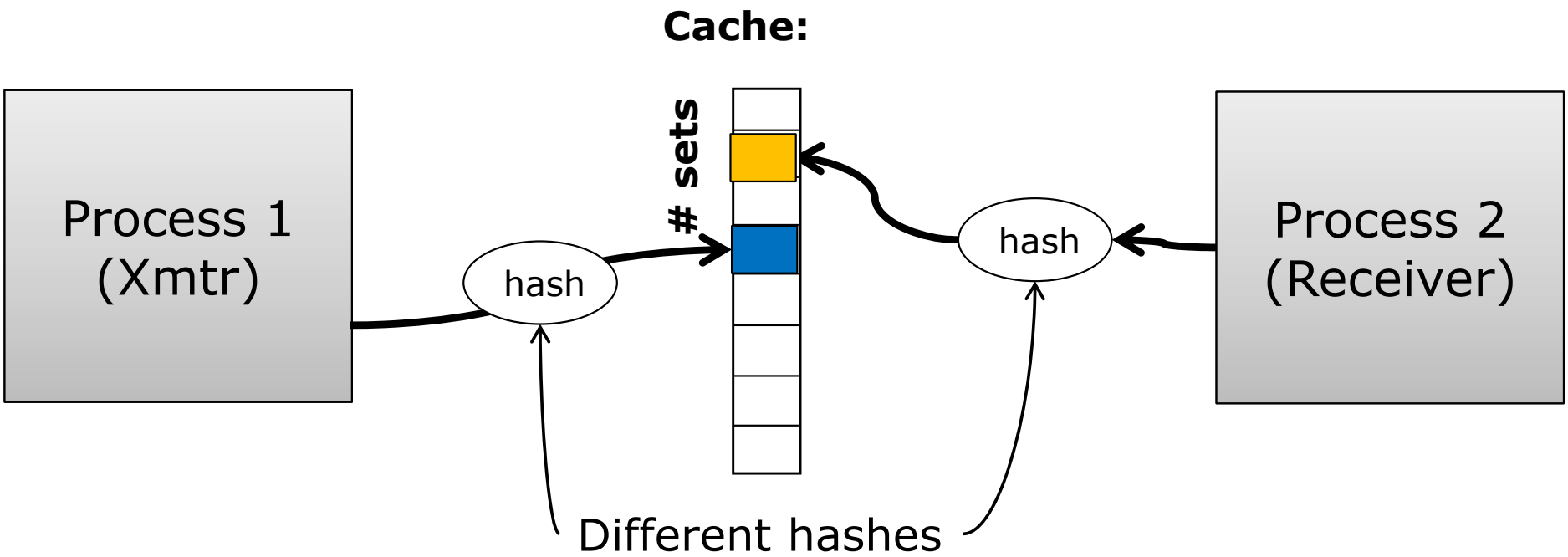
- Adding a single hash makes it difficult for the receiver to craft an address that monitors a specific set because addresses in each process will not match one-to-one.

Communication with subchannels



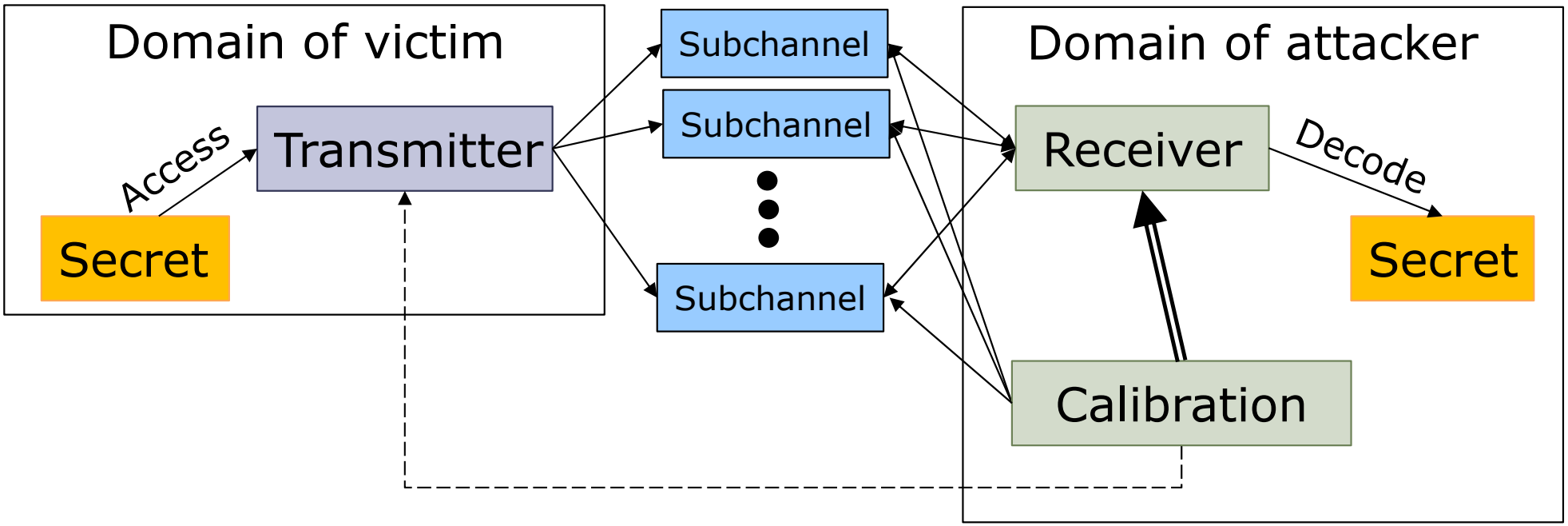
- Transmissions may now occur on one of many subchannels
- With a single hash, analysis by the receiver can, however, figure out (reverse engineer) which subchannel will be modulated.

Obfuscating the channel (2)



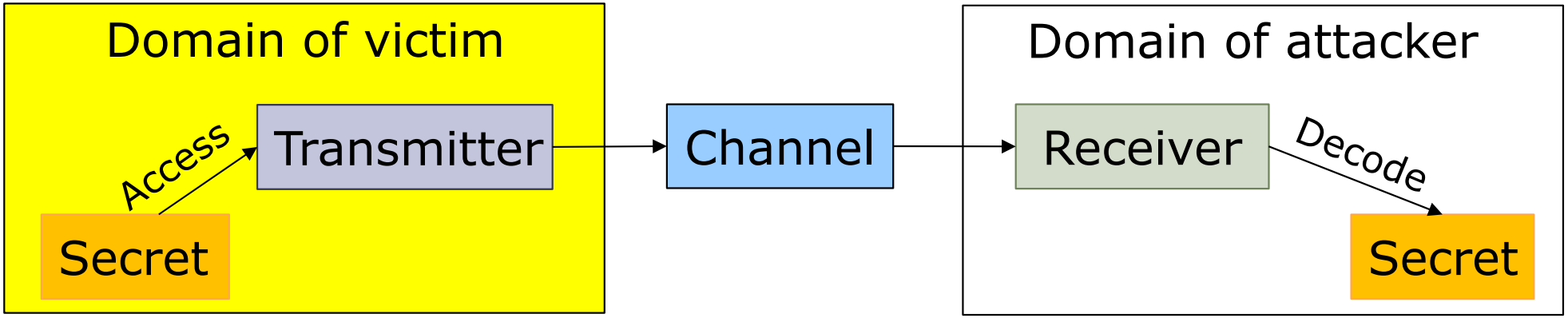
- Adding a process dependent hash makes the needed cache collision probabilistic.
- Now the receiver needs an extra step to find a way to probe a variety of "channels" to detect modulation.

Receiver Calibration



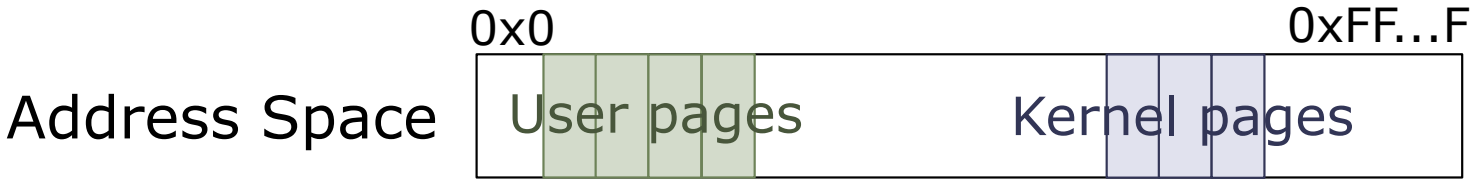
- The calibration unit determines which subchannels the receiver needs to use to detect modulation by a transmission
- During calibration, the receiver may just observe known transmissions by the transmitter or provoke the transmitter to make a particular transmission.

Types of Transmitters



- Types of transmitter:
  1. Pre-existing so victim itself leaks secret, (e.g., RSA keys)
  2. Programmed and invoked by attacker (e.g., Meltdown)

Reminder: Speculative Execution



- In x86, a page table can have kernel pages which are only accessible in kernel mode:
  - This avoids switching page tables on context switches, but
  - Hardware speculatively assumes that there will not be an illegal access, so instructions following an illegal instruction are executed speculatively.
- So what does the following code do when run in user mode do?

```
val = *kernel_address;
```

- Causes a protection fault, but data at "kernel_address" is speculatively read and loaded into val.

Meltdown [Lipp et al. 2018]

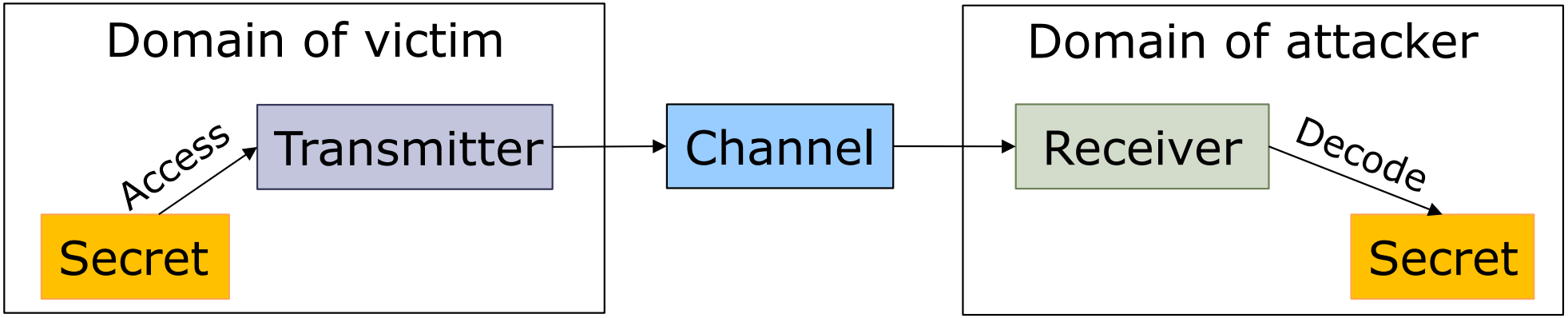
1. Preconditioning: Receiver allocates an array `subchannels[256]` and flushes all its cache lines
2. Transmit: Transmitter (controlled by attacker) executes

```
uint8_t secret = *kernel_address;
subchannels[secret] = 1;
```

3. Receive: After handling protection fault, receiver times accesses to all of `subchannels[256]`, finds the subchannel that was "modulated" to decode the secret.

- Result: Attacker can read arbitrary kernel data!
  - For higher performance, use transactional memory (protection fault aborts transaction on exception instead of invoking kernel)
  - Mitigation? Do not map kernel data in user page tables (KPTI)
    Return zero upon permission check failure (supporting precise exception)

Types of Transmitters



- Types of transmitter:
  1. Pre-existing so victim itself leaks secret, (e.g., RSA keys)
  2. Programmed and invoked by attacker (e.g., Meltdown)
  3. Synthesized from existing victim code and invoked by attacker (e.g., Spectre v2)

Spectre variant 1
[Kocher et al. 2018]

- Consider a situation where there is some kernel code that looks like the following:

```
xmit: uint8_t index = *kernel_address;
      uint8_t dummy = random_array[index];
```

- Interpret that code as a transmitter:

```
xmit: uint8_t secret = *kernel_address;
      uint8_t dummy = subchannels[secret];
```

- But this kernel code is protected by a branch. *Can we make the kernel speculatively execute "xmit"?*

```
if (kernel_address is public_region) {
   uint8_t index = *kernel_address;
   uint8_t dummy = subchannels[index];
}
```

Conditional branch misprediction

Spectre variant 1
[Kocher et al. 2018]

- Consider the following kernel code, e.g., in a system call

```
if (x < array1_size)
  y = array2[array1[x] * 4096];
```

1. Precondition: Flush all the elements in array2 from cache
2. Train: Attacker invokes this kernel code with small values of x to train the branch predictor to be taken
3. Transmit: Attacker invokes this code with an <u>out-of-bounds</u> x, so that `&array1[x]` points to a desired kernel address. Core mispredicts branch, <u>speculatively</u> fetches address `&array2[array1[x] * 4096]` into the cache.
4. Receive: Attacker probes cache to infer which line of `array2` was fetched, learns data at kernel address

Spectre variant 2
[Kocher et al. 2018]

- Can also exploit indirect branch predictor:
  - Most BTBs store partial tags for source addresses

Victim_branch

training_branch

```
        kernel_address = a_desired_address;
        jump some_where_else
        …
        kernel_address = a_safe_address;
        jump xmit
        …
xmit:   uint8_t secret = *kernel_address;
        uint8_t dummy = subchannels[secret];
```

1. Train: trigger `victim_branch -> xmit` many times
2. Transmit: '`victim_branch`' and '`training_branch`' alias in BTB, so we can speculatively trigger `victim_branch -> xmit`
3. Receive: similar to Spectre v1

Spectre variants and mitigations

- Spectre relies on speculative execution, not late exception handling → Much harder to fix than Meltdown
- Several other Spectre variants reported
  - Leveraging the speculative store buffer, return address stack, leaking privileged registers, etc.
- Can attack any type of VM, including OSs, VMMs, JavaScript engines in browsers, and the OS network stack (NetSpectre)
- Short-term mitigations:
  - Microcode updates (disable sharing of speculative state when possible)
  - OS and compiler patches to selectively avoid speculation
- Long-term mitigations:
  - Disabling speculation?
  - Closing side channels?

Summary

- ISA is a timing-independent interface, and
  - Specify *what* should happen, not *when*
- ISA only specifies architectural updates
  - *Micro-architectural changes are left unspecified*
- Implementation details (e.g., speculative execution) and timing behaviors (e.g., microarchitectural state, power, etc.) have been exploited to breach security mechanisms.
- ISA, as a software-hardware contract, is insufficient for reasoning about microarchitectural security

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Learn to attack processors…

Side channel attacks

Spectre, Meltdown, Foreshadow

Row-hammer attacks

Intel SGX

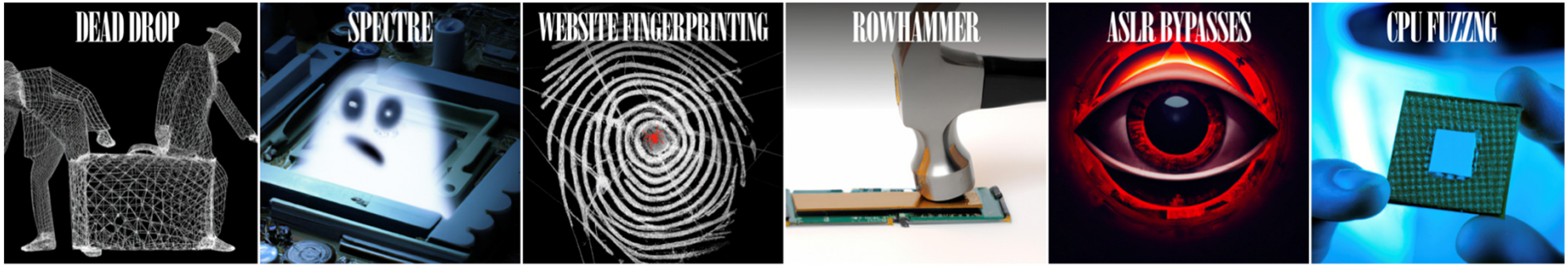
ARM TrustZone

Hardware mitigations for ROP/ JOP

And how to defend them!

Secure Hardware Design @ MIT

Making Computer Architecture Fun!



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Thank you!